



Name: \_\_\_\_\_  
 Class: \_\_\_\_\_  
 Date: \_\_\_\_\_

Mr. Rawson • IB DP Physics

# Simple Harmonic Motion 7 – Practice IB Questions

Paper 1A and Paper 2 style

## Before you start

Have your **calculator in radians** and your **IB data booklet** open. In every calculation, write the equation first, then substitute, then give the answer with a unit. **Check:**  $\sin(\pi/2) = 1$  on your calculator.

## The command terms on this sheet, as the IB defines them

|                  |                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>State</b>     | Give a specific name, value or other brief answer without explanation or calculation.                                                                                              |
| <b>Calculate</b> | Obtain a numerical answer showing the relevant stages in the working.                                                                                                              |
| <b>Show</b>      | Give the steps in a calculation or derivation. In <i>show that</i> , the answer is given: the marks are for the working, so give one more significant figure than the value shown. |
| <b>Determine</b> | Obtain the only possible answer.                                                                                                                                                   |
| <b>Explain</b>   | Give a detailed account including reasons or causes.                                                                                                                               |

## SL

Everyone answers this section.

### Multiple choice (Paper 1A style: circle one letter)

#### Question 1

1 mark

**Relevant equations:**  $\omega = 2\pi f$

An object oscillates with simple harmonic motion at a frequency of 2.5 Hz. What is the angular frequency of the oscillation?

- A. 0.40 rad/s      B. 2.5 rad/s      C. 7.9 rad/s      D. 16 rad/s

#### Question 2

1 mark

**Relevant equations:**  $T = 2\pi\sqrt{\frac{m}{k}}$      $v_{\max} = \omega x_0$      $a_{\max} = \omega^2 x_0$

A mass on a spring oscillates with simple harmonic motion. The amplitude is doubled. What happens to the period  $T$ , the maximum speed  $v_{\max}$  and the maximum acceleration  $a_{\max}$ ?

|    | $T$       | $v_{\max}$ | $a_{\max}$ |
|----|-----------|------------|------------|
| A. | unchanged | unchanged  | unchanged  |
| B. | unchanged | doubled    | doubled    |
| C. | doubled   | unchanged  | unchanged  |
| D. | doubled   | doubled    | quadrupled |

**Question 3****1 mark**

**Relevant equations:**  $v_{\max} = \omega x_0$     $a_{\max} = \omega^2 x_0$     $T = \frac{2\pi}{\omega}$

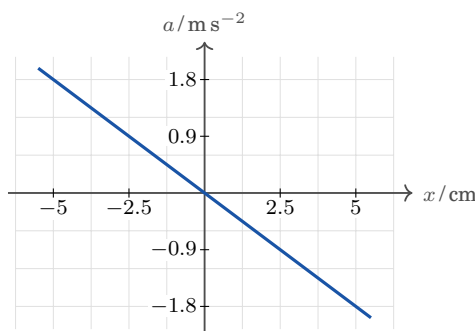
An oscillator in simple harmonic motion has a maximum speed of 0.60 m/s and a maximum acceleration of 3.0 m/s<sup>2</sup>. What is its period?

- A. 0.20 s                      B. 1.3 s                      C. 5.0 s                      D. 7.9 s

**Question 4****1 mark**

**Relevant equations:**  $a = -\omega^2 x$     $\omega = 2\pi f$

The graph shows how the acceleration  $a$  of an oscillating object varies with its displacement  $x$ .



What is the frequency of the oscillation?

- A. 0.95 Hz                      B. 5.7 Hz                      C. 6.0 Hz                      D. 36 Hz

**Structured questions** (Paper 2 style)**Question 5****6 marks**

**Relevant equations:**  $\omega = \frac{2\pi}{T}$     $v_{\max} = \omega x_0$     $a_{\max} = \omega^2 x_0$     $a = -\omega^2 x$

A car drives over a bump. Afterwards, the car body moves up and down on its springs with simple harmonic motion of period 0.80 s and amplitude 4.0 cm.

- (a) Calculate the angular frequency of the oscillation.

**[1]**


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- (b) Calculate the maximum speed of the car body.

**[1]**


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- (c) Calculate the maximum acceleration of the car body, and state where in the oscillation it occurs.

[2]

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- (d) The motion of the car body is described by  $a = -\omega^2 x$ . State **two** features of the acceleration that this equation describes.

[2]

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**Question 6**

**6 marks**

**Relevant equations:**  $T = 2\pi\sqrt{\frac{l}{g}}$     $\omega = \frac{2\pi}{T}$     $v_{\max} = \omega x_0$

A child sits on a playground swing. The swing behaves as a simple pendulum of length 2.5 m.

- (a) Show that the period of the swing is about 3.2 s.

[2]

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- (b) The seat moves 0.60 m either side of its equilibrium position. Calculate the maximum speed of the seat.

[2]

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- (c) A heavier child sits on the same swing. State and explain the effect on the period.

[2]

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**Relevant equations:**  $\omega = \sqrt{\frac{k}{m}}$     $T = \frac{2\pi}{\omega}$     $a = -\omega^2 x$

A trolley of mass 0.40 kg sits between two springs on a level track. When it is displaced by  $x$  from its equilibrium position, the resultant force on it is  $F = -36x$ , with  $F$  in newtons and  $x$  in metres.

- (a) Explain why the trolley performs simple harmonic motion when it is released.

[2]

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- (b) Calculate the period of the oscillation.

[2]

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HL students: answer this section as well. It uses the HL equations from your data booklet.

### Multiple choice (Paper 1A style: circle one letter)

#### Question 8

1 mark

**Relevant equations:**  $x = x_0 \sin \omega t$     $\omega = \frac{2\pi}{T}$

An object moves in simple harmonic motion with period  $T$  and amplitude  $x_0$ . It passes through the equilibrium position at  $t = 0$ . What is the shortest time for it to reach a displacement of  $\frac{1}{2}x_0$ ?

- A.  $\frac{T}{12}$                       B.  $\frac{T}{8}$                       C.  $\frac{T}{6}$                       D.  $\frac{T}{4}$

#### Question 9

1 mark

**Relevant equations:**  $x = x_0 \sin(\omega t + \phi)$     $v = \omega x_0 \cos(\omega t + \phi)$

The displacement of an oscillator is given by  $x = x_0 \sin(\omega t + \phi)$ . At  $t = 0$  the oscillator passes through the equilibrium position moving in the negative direction. What is  $\phi$ ?

- A. 0                      B.  $\frac{\pi}{2}$                       C.  $\pi$                       D.  $\frac{3\pi}{2}$

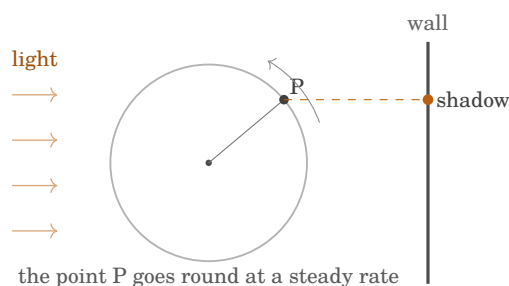
### Structured questions (Paper 2 style)

#### Question 10

7 marks

**Relevant equations:**  $\omega = 2\pi f$     $x = x_0 \sin \omega t$     $v = \omega x_0 \cos \omega t$     $v = \pm \omega \sqrt{x_0^2 - x^2}$

A small peg P is fixed to the edge of a turntable, 0.12 m from its centre. The turntable rotates at a steady 45 revolutions per minute. Light shines from the side, so the shadow of P on a wall moves up and down with simple harmonic motion.



- (a) Show that the angular frequency of the shadow's motion is about 4.7 rad/s.

[1]

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- (b) At  $t = 0$  the shadow passes through its equilibrium position, moving upwards. Calculate the displacement of the shadow at  $t = 0.20$  s.

[2]

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- (c) Calculate the velocity of the shadow at  $t = 0.20$  s.

[2]

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- (d) Use your answer to (b) to confirm your answer to (c).

[1]

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- (e) A second peg is fixed to the edge of the same turntable, a quarter of a turn ahead of P. State the phase difference between the motions of the two shadows.

[1]

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